

Braillix - Build Pack

Written 2026-08-15. For the 24 August evaluation. Scope: **electronics, assembly and sourcing**. Mechanical/CAD issues are *flagged* here and fixed in the CAD fork, never here.

This is the single document to work from at the bench. It answers, in order: what to present on 24 August, what can be permanently assembled now, what goes in which box, how every wire connects, why each part exists, and exactly what is still to be ordered.

PART 0 - The 24 August plan

The upper linkages are not on the critical path

The six linkages are the hardest parts in the project and they are giving trouble. **Do not let them decide whether you have a demo.** Nothing in the electronics or the actuation chain depends on them existing.

Split the project the way an evaluator will accept:

Layer	State on 24 Aug	How it is shown
Encoding - 64 states, 6 dots	[OK] Solved, provable	Simulator + the 64-state table
Actuation - motor, cam, homing	[OK] Buildable now	Real hardware turning on the bench
Control - ESP32, WiFi dashboard	[OK] Buildable now	Phone browser, live
Transmission - linkages to dots	[!] In progress	Simulator, plus one linkage held by hand

Three of four layers are real hardware. That is a legitimate mid-project result, and it is far stronger than a fully-assembled unit that does not move.

What the demo actually looks like

```
[ Phone ] --wifi--> [ ESP32 ] --> [ ULN2003 ] --> [ 28BYJ-48 ] --> [ CAM DISC ]
type "h"           picks state  switches      turns 5.625deg  visibly
                   17 of 64    the coils      per state   rotates
                               ^
                               [ HALL SENSOR ] finds home on power-up
```

Projected beside it: the 3D simulator, showing the SAME state number driving the six dots up and down.

Say this out loud to the panel: the cam is an absolute 6-bit rotary encoder; the hardware proves it reaches every one of 64 positions repeatably; the simulator proves each position lifts the correct dots. **The linkage is the assembly step in progress.** That framing is honest and it is the actual engineering status.

[!] **Do not promise raised dots on 24 August.** If the linkages come good before then, it is a bonus you add. Building the demo around them is how you end up with nothing.

PART 1 - What is safe to make PERMANENT right now

You have the iron. The question is what you can commit to without regretting it.

#	Job	Permanent?	Verdict
1	Header pins onto the DC jack's bare wires	Yes	[OK] Do it. The jack's job never changes.
2	Wires soldered flat to ULN2003 IN1-IN4 + power	Yes	[OK] Do it. Required by the 14mm pocket, and the ULN2003 end of those wires is fixed no matter which GPIO the other end lands on.
3	The power spine (one 5V source, three branches)	Yes	[OK] Do it. Wire it with a screw terminal block if you would rather not splice.
4	Heat-set brass inserts into the printed parts	Yes	[OK] Do it - see Part 7. The iron you just bought is the tool for this.

#	Job	Permanent?	Verdict
5	Desoldering the hall TO-92 off its blue module	Yes	[!] Wait. See the caution below.
6	Super glue on the dot_insert tile	Yes	[!] Wait until the top plate and the dot alignment are confirmed.
7	Soldering anything directly to ESP32 GPIO pins	Yes	[!!] Never. See below.

[!!] Rule: nothing is soldered to an ESP32 GPIO

The pin map is not frozen and will change twice more:

1 The GPIO21/22 conflict fix - IN3 moves to **23**, IN4 to **27** (ELECTRONICS_PLAN.md Part 3).

1 The expander migration when a third cell appears (Part 5 of the same doc).

Every ESP32 connection stays **Dupont jumper into the header**. Solder is for the far end of the wire, never the board end. This costs nothing and keeps the board reusable.

[!] Caution before desoldering the hall sensor

The blue MH module is what makes the sensor usable today. Taking the sensor off it is **irreversible and only needed for final assembly in the pod** - not for the demo, where the module sits on the breadboard.

[!] **Read the part number on the black 3-legged sensor first.**

Marking	Type	If you desolder it
49E / SS49	Analog	Fine - outputs a voltage directly, works on A0 with no extra parts
3144 / A3144	Digital, open-collector	[!] Needs a 10k? pull-up to 3V3 that currently lives on the module. Without it the output reads nothing.

Do not desolder before 24 August. There is no benefit and a real chance of losing your only sensor.

PART 2 - Which box holds what

<pre> +=====+ BRAIN POD (one, ever) +=====+ ESP32 DevKit 51x28mm 3 nav buttons DC power jack (3+ cells only: 2x 4.7k pull-ups) +=====+ </pre>	<pre> +=====+ MUSCLE CELL (one per character) +=====+ 28BYJ-48 motor 19mm tall Cam disc ULN2003 board [!] 14mm max Hall sensor (bare TO-92) (3+ cells only: MCP23017) +=====+ </pre>
<pre> +----- 5V, GND, 4 control, 1 sense --+ (fixed 4 wires once expanders arrive) </pre>	<pre> ^ </pre>

Vertical stack inside a cell - where the millimetres go

z=58	+-----+	top plate top surface, braille dots
z=54	+-----+	top plate underside
	8mm standoffs	
z=45	-----	cam flat surface (+0.8 at a bump)
z=43	+-----+	cam disc, 2mm floor
z=41	+-----+	base plate top / motor face
	19mm 28BYJ-48 body	
z=22	+-----+	
z=20	+-----+	mid plate, 2mm
	[!] 14mm ELECTRONICS POCKET <-- the only tight space	
z=4	+-----+	
z=0	+-----+	box floor, 4mm

Space, honestly

Location	Space	Occupant	Margin
Electronics pocket	36 x 46 x 14mm	ULN2003, wires soldered flat ~12mm	[!!] ~2mm. Zero slack.
Pod interior	~38mm headroom	ESP32 + jack + buttons	[OK] Wildly oversized
Cam-to-top-plate gap	8mm standoffs	linkages, 12.2mm total height	[!] Set by the CAD fork

[!!] **The 14mm pocket is why soldering exists in this project at all.** A ULN2003 with Dupont jumpers plugged in vertically stands ~20mm and will not close the box. Cut the connectors off and lay the wires flat and it drops to ~12mm.

[!] **The pocket lost 2mm** (16 -> 14) in the 2026-08-01 CAD fix that corrected the motor stack. It still clears, but check the fit with real soldered wires before closing the box.

PART 3 - Every connection, colour coded

3.1 Power - do this first, and get it right

From	Wire	To	Note
5V adapter barrel	-	DC pigtail jack	
Jack RED (+5V)	RED	Breadboard + rail	[OK] Polarity measured correct 2026-08-01
Jack BLACK (GND)	BLACK	Breadboard - rail	
Breadboard + rail	RED	ULN2003 + pin	Motor power - 5V, not 3V3
Breadboard - rail	BLACK	ULN2003 - pin	
ESP32 GND	BLACK	Breadboard - rail	[!!] COMMON GROUND - see below

[!!] The two power rules that protect the hardware

1 Common ground is mandatory. The ESP32 runs off USB, the motor off the adapter. If their grounds are not joined, the hall sensor reads garbage and the motor behaves randomly. This is the single most common cause of "everything worked, now nothing".

1 Never connect the adapter to ESP32 VIN while USB is plugged into the laptop. Two supplies fighting each other back-feeds your laptop's USB port. Motor from the adapter, ESP32 from USB, **grounds joined only**.

3.2 Control - ESP32 to ULN2003

Use the **corrected** pins. Do not use 21/22.

ESP32	Wire	ULN2003	Why this pin
GPI018	ORANGE	IN1	Safe output
GPI019	YELLOW	IN2	Safe output
GPI023	GREEN	IN3	[!] Was 21 - frees I2C SDA
GPI027	BLUE	IN4	[!] Was 22 - frees I2C SCL

[!!] **Never use GPIO 0, 2, 5, 12, 15 for a motor coil.** They are *strapping pins*, sampled at boot to decide how the chip starts. A coil pulling one of them low can stop the ESP32 booting, and it looks like a dead board.

3.3 Sense - the hall sensor

ESP32	Wire	Hall module	Caution
3V3	RED	VCC	[!!] 3V3, NEVER 5V. ESP32 GPIOs are not 5V-tolerant. A0 would output up to 5V and damage the pin.
GND	BLACK	GND	

ESP32	Wire	Hall module	Caution
GPI034	WHITE	A0	Analog. Input-only pin on ADC1 - correct choice.
-	-	D0	Unused for one cell. Needed from 3 cells up (expanders have no ADC).

3.4 Motor - the 28BYJ-48 plug

The motor's white 5-pin plug goes into the white socket on the ULN2003 board. **It only fits one way**. Colours are listed for diagnosis only:

Wire	Role
RED	Common / centre tap -> +5V
ORANGE	Coil 1
YELLOW	Coil 2
PINK	Coil 3
BLUE	Coil 4

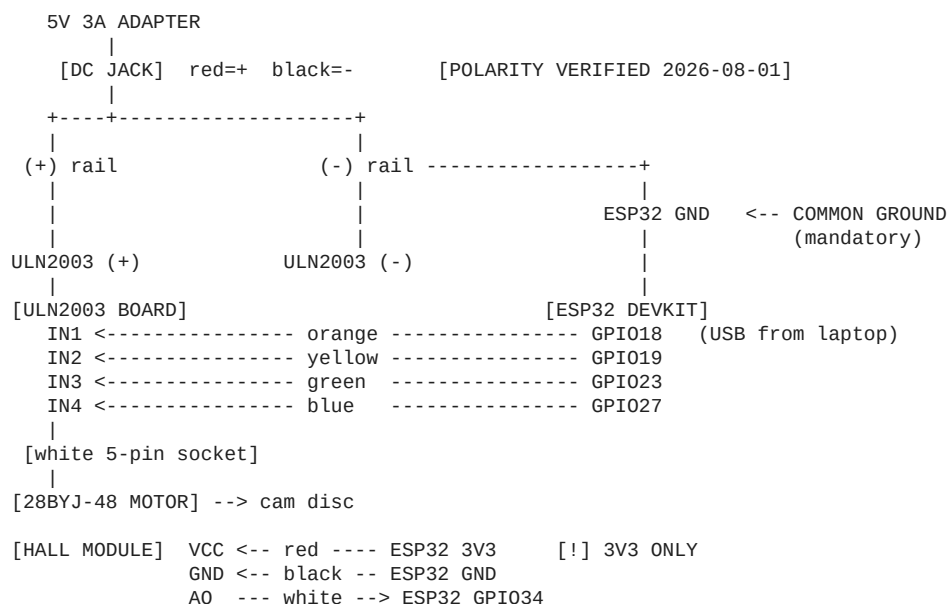
[!] If the motor buzzes and vibrates but does not turn, the coil order is wrong, not the wiring. Swap IN2 and IN3 in firmware before touching any hardware.

3.5 Navigation buttons - brain pod

ESP32	To	Config
GPI032	Button "Prev" -> GND	INPUT_PULLUP, active LOW
GPI025	Button "Select" -> GND	INPUT_PULLUP, active LOW
GPI017	Button "Next" -> GND	INPUT_PULLUP, active LOW

No resistors needed. The ESP32 has internal pull-ups; the button only has to pull the pin down to ground.

3.6 The whole circuit on one page



PART 4 - Why each component exists

Nothing here is decorative. One line each, so you can answer the panel.

Component	Why it exists	What breaks without it
ESP32 DevKit	Decides *which* of the 64 cam states to show; serves the WiFi dashboard	No control at all
ULN2003 board	An ESP32 pin gives ~20mA at 3.3V; a motor coil wants ~200mA at 5V. This chip is seven electronic switches that bridge that gap.	Connect the motor direct and you destroy the ESP32 pin
28BYJ-48 stepper	Moves in exact, countable steps - 4096 per turn. A DC motor cannot hold a position.	No repeatable positioning
Cam disc	The mechanical decoder: one angle = one 6-dot pattern. Six tracks read simultaneously.	6 separate actuators instead of 1
Hall sensor	Finds absolute zero on power-up. A stepper knows *relative* steps, never where it actually is.	Every character wrong after a power cut
Homing magnet (3x1mm)	The thing the hall sensor sees	Nothing to home against
Return springs	Push each linkage back down onto the cam. Gravity alone is too weak at this scale.	Dots stick up and never retract
DC jack + 5V 3A adapter	USB alone cannot supply motor current safely	Brown-outs, ESP32 resets mid-move
Brass heat-set inserts	PETG threads strip after a few open/close cycles. Brass does not.	The box becomes single-use
MCP23017 *(3+ cells only)*	Turns 5 wires per cell into a shared 4-wire bus	Pin exhaustion at ~4 cells

PART 5 - Cautions and precautions

Electrical

#	Rule	Consequence if broken
1	[!!] Hall VCC to 3V3 , never 5V	ESP32 input pin damaged
2	[!!] Common ground between the two supplies	Random behaviour, garbage sensor readings
3	[!!] Never adapter -> VIN while USB is connected	Back-feeds the laptop USB port
4	[!] Never a motor coil on GPIO 0, 2, 5, 12, 15	ESP32 will not boot
5	[!] De-energise coils when idle	Motor runs hot, wastes ~250mA holding still
6	[!] Power off before rewiring anything	Shorts happen while probing live

Soldering

#	Rule	Why
1	Heat the joint; let the joint melt the solder	Solder dabbed off the iron makes a cold joint - looks fine, fails later
2	Slide heat-shrink on before soldering	Everyone forgets this exactly once
3	Tug-test every joint	A good joint does not pull off. No glue rescues a bad one.
4	Practice on scrap first	Do not let the ESP32 be your first ever joint
5	[!] Keep a separate tip for heat-set inserts	Melted plastic and brass ruin a tip for solder work

Mechanical / space

#	Caution
1	[!!] 14mm pocket, ULN2003 at ~12mm. Dry-fit with real soldered wires before closing the box.

#	Caution
2	[!] Superglue on the dot_insert gives you no repositioning time . Tape-align it dry first, then wick glue in from the edge.
3	[!] Keep every adhesive away from the cam, the linkage feet and the dot holes. A drop of glue in the mechanism is worse than a bad solder joint.
4	[!] FDM holes print 0.2-0.3mm undersize. Expect to drill pilots to size.

PART 6 - Shopping: what is left

[OK] Already bought

Soldering iron - solder wire - wire stripper - super glue - multimeter - digital calipers - ESP32 - 28BYJ-48 + ULN2003 - hall module - 5V 3A adapter + jack - magnets 8x1mm - tactile switches - hookup wire - Dupont jumpers

6.1 Electronics still to buy

#	Item	Spec	Why	Est. ?	Where
1	Breadboard	830 tie points	[!!] The whole demo runs on it. This is the blocker.	100	Local / online
2	USB-C DATA cable	Must carry data	Charge-only cables look identical and fail silently. Test yours first.	150	Have one?
3	Heat-shrink assortment	2-6mm	Every joint. Non-negotiable.	100	Local
4	Hot glue gun + sticks	20-40W	Strain relief. Reworkable - this is why it beats epoxy while the pin map can still change.	250	Local
5	Flux paste	Rosin, no-clean	Makes ULN2003 pins far easier	100	Local
6	Spare iron tip	Any, cheap	[!] Dedicated to heat-set inserts	100	Local
7	Brass wool tip cleaner	-	Keeps the tip alive	100	Local
8	2-way screw terminal blocks	5mm pitch	Power spine without splicing	40	Local

Electronics subtotal: ~?940 (~?790 if your USB-C cable does data)

6.2 What NOT to buy yet

Item	Verdict
MCP23017 expander	[X] Not for 24 August, and not for two cells. It earns its place at 3+ cells only. Order it when you commit to cell 3 - Robu.in, Element14 India, or Amazon. Local shops are hit-and-miss for DIP-28 logic; that is not a problem you have to solve now.
Resistors, capacitors, diodes	[X] Zero needed. Flyback diodes are inside the ULN2003, pull-ups are inside the ESP32, regulators are on the DevKit.
ATmega328P / custom PCB	[X] Cancelled. See ELECTRONICS_PLAN.md Part 1.
Level shifters	[X] Hall runs at 3V3, so GPIO34 never sees 5V.

6.3 Mechanism parts (order with the screws)

#	Item	Spec	Qty	Est. ?
1	Micro compression springs	2.0mm OD , ~0.3mm wire, ~4mm free length	6 + spares	500
2	Homing magnet	[!!] 3mm dia x 1mm thick neodymium	buy 10	120

[!!] **Not pen springs.** Pen springs are ~4mm OD; braille rows are 2.6mm apart. They physically cannot fit. [!!] **Not 3x2mm magnets.** The cam floor is only 2mm - a 2mm-thick magnet's pocket cuts clean through it and destroys three cam tracks. 1mm thick leaves 0.8mm of floor and holds fine. Your 8x1mm magnets are too wide in diameter for this job (they are for docking).

PART 7 - Screws and inserts

7.1 The change that matters: brass heat-set inserts

The 2026-08-01 CAD update moved the box from *screwing into plastic* to **brass heat-set inserts**. PETG threads strip after a few open/close cycles, and this box gets opened constantly.

You install them with the soldering iron you just bought. Set ~230degC, press the insert in slowly and straight, let it cool before removing the iron.

[!] **Use a cheap spare tip.** Brass and melted plastic contaminate a tip permanently for soldering work.

7.2 What to order

Sizes taken from the CAD parameters, not guessed:

#	Part	Size	Qty	CAD source
1	Bolt - top plate -> standoffs -> box	M2.5 x 25mm , socket cap or button head	4	top_plate.scad screw_dia=3.2 clearance
2	Heat-set insert - box corner posts	M2.5 , OD 3.5mm , length 5.0mm	4 (buy 10)	outer_box.scad insert_m25_dia=3.5, depth=5.5
3	Screw - motor ears	M4 x 10mm , thread-forming preferred	2	base_plate.scad moto r_mount_pilot=3.3
4	Screw - pod lid	M2 x 8mm *(or M2.5 x 8 - see 7.3)*	2	esp32_pod_params.scad
5	Heat-set insert - pod lid posts	M2 , OD 3.2mm , length 4.0mm *(or M2.5 - see 7.3)*	2 (buy 10)	insert_m2_dia=3.2, depth=4.5

[!] **Measure the inserts when they arrive.** The CAD comments say so explicitly - insert OD varies by brand, and the bore is a parameter precisely so it can be adjusted. If your M2.5 inserts measure 3.6mm rather than 3.5mm, that is a one-line CAD change, not a reprint decision.

Search terms for onliscrow.com: "M2.5 socket cap screw 25mm", "brass threaded insert M2.5", "M4 socket cap 10mm". If inserts are not stocked there, they are reliably available on Robu.in and Amazon as "brass heat set insert assortment kit M2 M2.5 M3" (~\$350 for a mixed kit, which also covers you for the measuring surprise above).

[!] I cannot check live stock on any of these sites. Treat the list as *what to search for*, and if a size is genuinely unavailable, Part 7.3 tells you what can move.

7.3 Can we use one size for everything?

Almost - you can get from three sizes to two. Here is the honest analysis:

Location	Now	Can it change?	Reasoning
Box corners	M2.5 x 25	? Keep	Inserts and bores already in the CAD; changing means re-printing the box

Location	Now	Can it change?	Reasoning
Pod lid	M2 x 8	[OK] -> M2.5 x 8	An M2.5 insert needs a 3.5mm bore. The post is 6.5mm dia, leaving 1.5mm of wall - thinner than the 1.65mm it has now, but still enough for hot brass.
Motor ears	M4 x 10	[!!] Keep M4	The 28BYJ-48's own mounting ears are drilled 4.2mm . An M2.5 screw leaves 1.7mm of slop, and the motor would shift under load - which moves the cam off centre. This is the one place slop is unacceptable.

Recommendation

Standardise on M2.5 for everything structural. Keep M4 x 2 for the motor only. Three sizes become two, and you buy one insert size instead of two.

CAD change required - flagged for the CAD fork, NOT applied here:

```
cad/scad/esp32_pod_params.scad
insert_m2_dia    3.2  ->  3.5    M2.5 insert bore
insert_m2_depth  4.5  ->  5.5    M2.5 inserts are longer
lid_clearance_hole 2.2 ->  2.9    M2.5 clearance
```

[!] **Only worth doing if the pod shell and lid are not yet printed.** If they are printed, buying two M2 screws is far cheaper than a reprint. **Check before changing anything.**

PART 8 - Build order for the next two weeks

#	Step	Needs	Blocked by
1	Build the breadboard circuit	Breadboard, Dupont	[!!] Only the breadboard. No soldering.
2	Flash breadboard_test.ino	Data USB cable	Step 1
3	Confirm motor turns + hall homes + dashboard loads	-	Step 2
4	Solder header pins to the DC jack	Iron [OK]	Nothing
5	Solder ULN2003 wires flat	Iron [OK], heat-shrink	Nothing
6	Install brass inserts	Iron [OK], spare tip, inserts	Ordering
7	Dry-fit ULN2003 in the 14mm pocket	Printed box	Step 5
8	Rehearse the demo script	-	Step 3

[!!] **Step 1 is the whole project's blocker and it needs nothing you do not already own except a breadboard.** Everything else on this list is polish.

PART 9 - Flagged for the CAD fork (do not fix here)

- Pod lid fastener standardisation** - M2 -> M2.5, three parameters. Only if not yet printed.

See Part 7.3.

- Hall module does not fit** the base plate - needs the bare TO-92 desoldered and a pocket that holds it. [!] Check the sensor part number first (Part 1).

- DC jack has no thread or nut** - the pod lid cannot clamp it. Needs a retainer or a different jack.

- Electronics pocket is 14mm** with a ~12mm occupant. Working, zero margin. Verify with real soldered wires.

- Upper linkages** - the 24 August risk. Electronics does not depend on them.